UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	B	How many times does a while loop execute when its condition is false initially?	while is a pretest loop; a false condition skips the body.
A2	B	Which values can a bool variable hold?	A bool holds true or false.
A3	C	Which tool represents an algorithm using standard graphical symbols?	A flowchart uses standard symbols such as terminals, processes and decisions.
A4	A	Which loop is commonly used to process every array element?	A for loop with an index visits each element.
A5	C	Which term describes a finite sequence of precise steps that solves a problem?	An algorithm is a finite, ordered set of unambiguous steps.
A6	C	Which keyword prevents a variable's value from being changed after initialisation?	const defines a read-only variable.
A7	C	Which statement immediately exits a loop?	break terminates the nearest loop or switch.
A8	A	What does the % operator return?	% is the modulus operator and returns the remainder.
A9	A	Which input pattern stops when the user enters -1?	A sentinel value ends the loop.
A10	A	Which loop is guaranteed to execute its body at least once?	do...while tests the condition after the body.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: fundamentals

a) Define syntax error and logic error.

A syntax error violates the language grammar and is detected by the compiler. A logic error produces incorrect output although the program runs.

Marking: Award 1 mark for each definition.

(2 marks)

b) Explain TWO advantages of high-level languages over machine language.

They are easier to read and write, portable across machines, and supported by compilers and debugging tools.

Marking: Award 1.5 marks for each valid advantage.

(3 marks)

QUESTION 2

Topic: fundamentals

a) Draw a flowchart for the sum and average calculation.

Start -> read N -> total = 0, count = 1 -> while count <= N add count -> average = total / N -> display total and average -> end.

Marking: Award marks for input, initialisation, loop condition, accumulator, average and output.

(8 marks)

b) Write pseudocode and a complete C++ program for the algorithm.

Use a for loop from 1 to N, accumulate total, compute average as double(total)/N, then display both values.

Marking: Award marks for loop, accumulator, correct average type and output.

(5 marks)

QUESTION 3

Topic: data types

a) Evaluate the following expressions: (i) 7 + 3 * 2, (ii) (7 + 3) * 2, (iii) 17 % 5, (iv) 17 / 5, (v) 17.0 / 5.

(i) 13, (ii) 20, (iii) 2, (iv) 3, (v) 3.4

Marking: Award 1 mark per correct value.

(5 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

int a, b; cin >> a >> b; cout << a+b << a-b << a*b << a%b;

Marking: Award marks for declarations, input, four operations and output.

(6 marks)

QUESTION 4

Topic: input output

a) Trace the program and state the output.

30

Marking: The sum of 2+4+6+8+10 is 30.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

int i=0,s=0; do { if(a[i]>5) s+=a[i]; i++; } while(i<5); cout<<s;

Marking: Award marks for do...while, condition, filter and output.

(9 marks)

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QUESTION 5

Topic: arrays

a) Declare two parallel arrays for five sports and initialise them with suitable values.

string sports[5] = {"Swimming","Badminton","Athletics","Cycling","Hockey"}; int medals[5] = {8,12,7,5,9};

Marking: Award marks for both declarations, correct sizes and related indices.

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

Loop i=0..4 to display sports[i] and medals[i], accumulate total, average = double(total)/5.

Marking: Award marks for paired display, total, average and output.

(8 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
```

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```
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

END OF MARKING SCHEME

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